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Renewable Energy

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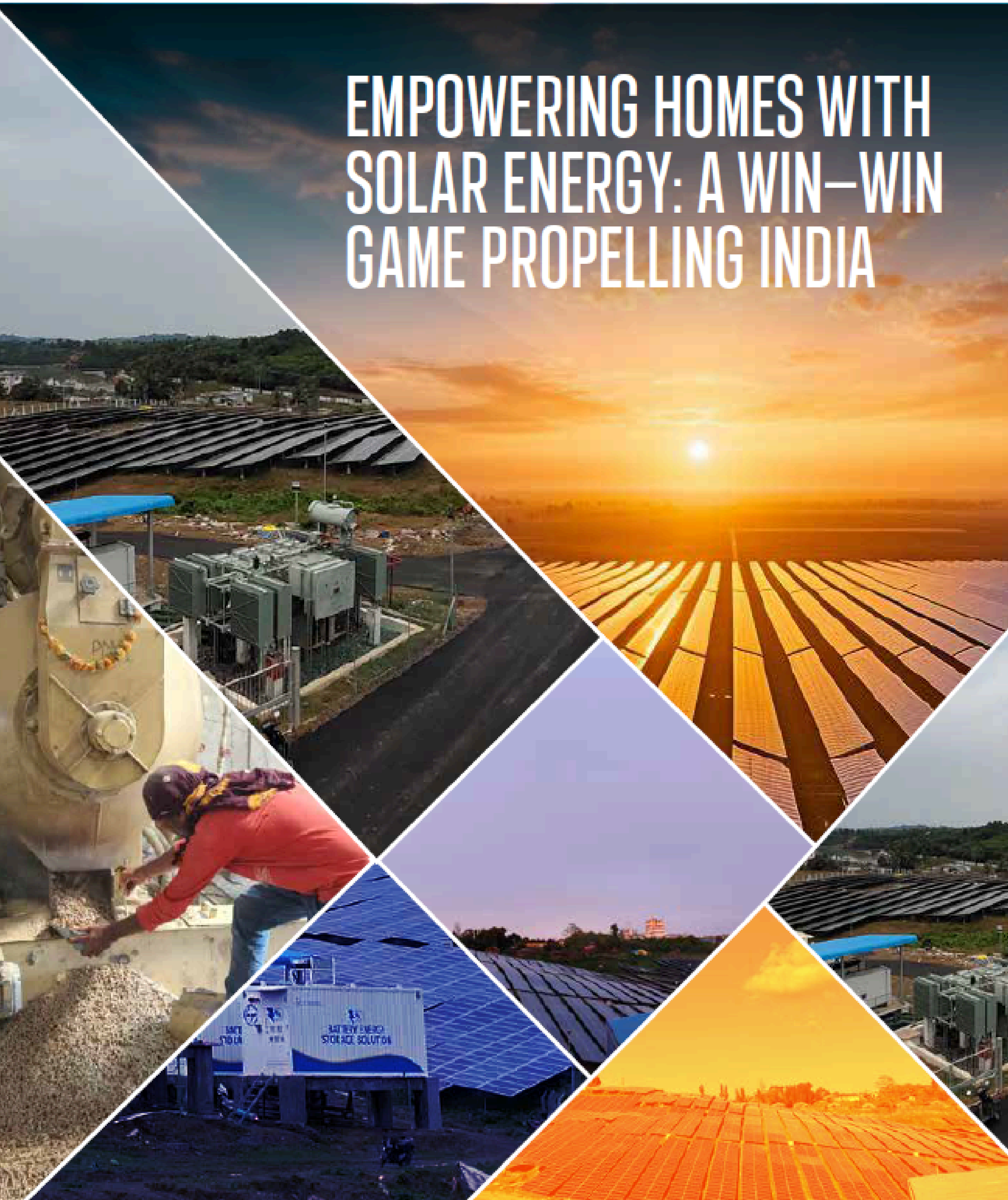
EMPOWERING HOMES WITH SOLAR ENERGY: A WIN–WIN GAME PROPELLING INDIA



Agriculture Sector Set to Provide Energy Too

Opportunities waiting to be tapped gainfully

The waste of one sector becomes the input for another. Agricultural residues, if used effectively, can fuel thermal power plants to generate electricity. They can literally move the transportation sector too through Bio-CNG. Well, it's a serious business opportunity! Christian Rakos and Sunil Dhingra give you some insights...

India has 197 million hectares of gross land area under crop cultivation with a large variety of crops being grown. According to a study by MNRE, India generates about 774 million tonnes (MT) of crop residue annually. The yearly surplus crop residues amount to approximately 228 MT, much of which is currently burned in the fields causing heavy air pollution. As a matter of fact, burning this crop residue is akin to wasting a valuable renewable energy source.

Two major opportunities have emerged to utilize agricultural residues in a more meaningful way: (a) the use of densified solid biomass to replace coal and other fuels used for electricity and industrial heat generation and (b) the production of compressed methane from anaerobic fermentation of biomass and organic waste materials, usually referred to as Bio-CNG. These two opportunities are briefly discussed below.

Figure 1. Crop residues collection and aggregation in field

Co-firing of biomass in thermal power stations

The Ministry of Power in its National Mission on Biomass Cofiring has mandated thermal power plants (TPPs) to replace 5% of coal with biomass by FY 2023/24. What seems a modest call is an enormous challenge, as it translates into utilizing approximately 37 MT of biomass, mostly in the form of pellets to replace coal. Global pellet production is currently around 60 MT. By FY 2025/26, this mandate will be increased to 7% of coal replacement, which would increase the demand for biomass by 40%. A benchmark price of about Rs 8000 per tonne of pelletized straw has been published by the Ministry of Power. The price varies slightly depending on the region. In all, 65 TPPs started co-firing with cumulative biomass co-firing 11.16 lakh toones generating 1408 million units (MU) of electricity (as on 15 October 2024). This biomass co-firing has resulted in avoiding CO_2 emissions to the tune of 13.4 lakh toones cumulatively.

Figure 2: Agro-residues based pellet production plant

Production of Bio-CNG

The use of agricultural as well as urban and industrial wastes of organic origin for producing biogas and compressed methane (Bio-CNG) for the transport sector presents itself as a big opportunity. MNRE supports the development of this technology too as part of its Waste to Energy Program. This program has a budget allocation of Rs 600 crore from 2021/22 to FY 2025/26. Two other programs – the SATAT program and the GOBARdhan program – have also supported the establishment of Bio-CNG plants.

The Indian biogas market size was valued at Rs 10,279 crore in 2021 and is expected to grow to Rs 16,494 crore by 2029, at a compound annual growth rate (CAGR) of 6.3%. India is the world's second-largest biogas consumer in the world. According to the IBA (Indian Biogas Association), India creates over 2.5 lakh tonnes of trash daily, and over 85% of all organic waste generated presently ends up in landfills.

India generates 0.4 kg of waste per capita, roughly half of which is organic and can be transported to CBG facilities. In addition, agricultural wastes and wastes from the processing of crops are attractive feedstocks for CBG production. At present, 68 CBG plants have been commissioned and about 22,097 tonnes of CBG has been sold in 2023/24. A 5% CBG mandate was introduced in the Union Budget 2023/24. Excise duty on CBG has been exempted too. Besides, the government guarantees a minimum price of Rs 54 per kg (plus GST) for CBG upon meeting specified standards.


Figure 1. Agriculture waste to CBG production plant

The MNRE has implemented the bio-energy program and has been promoting the use of biomass for energy generation (See Box 1).

Challenges

While both these technologies hold great promises, they also come with a set of challenges. A shared challenge is the need to collect agricultural residues in a fairly short period of time. Very large volumes need to be stored, protected from rain and gradually processed to be usable as fuel. Solving these logistic challenges is a precondition for using biomass extensively as a source of energy.

Technically, pelletization of agro-residues is less common in India than biogas production and it is likely, that the challenges of running a pellet plant successfully are underestimated. These challenges include (a) complex technical issues, (b) risks related to fire and dust explosion incidents, and (c) working hazards. The MNRE is currently in the process of forging a partnership with the World Bioenergy Association to support the transfer of knowhow to the nascent Indian pellet industry.

Economically, in both cases, the challenge is to compete with fossil fuels in terms of price, logistics, and supply chain management.

Which way to go?

Given the fact that both the direct use as fuel for combustion and the production of methane as transport fuel are attractive, the question could be asked – well, which route to take best? The answer to this question depends on many factors.

Box 1. MNRE's Biogas Program: specific components

- Setting up of biogas plants for clean cooking fuel, lighting, meeting thermal and small power needs of users which results in GHG reduction, improved sanitation, women empowerment, and creation of rural employment.
- For production of organic enriched bio-manure: The digested slurry from biogas plants, a rich source of manure, shall benefit farmers in supplementing or reducing the use of chemical fertilizers.
- To promote biogas-based decentralized renewable energy sources of power generation (off-grid), in the capacity range of 3–250 kW or thermal energy for heating/cooling applications from the biogas generation produced from biogas plants of size 25–2500 m^3.

Scheme to Support Manufacturing of Briquettes and Pellets and Promotion of Biomass (non-bagasse) based Cogeneration in Industries

- The objective of the program is to support setting up of biomass briquette/pellet manufacturing plants and to support biomass (non-bagasse) based cogeneration projects in industries in the country. The scheme supports the development of biomass pelletization and torrefaction plants with financial incentives ranging from up to Rs 45 lakh to Rs 2.1 crore per project depending on the technology the facility uses.

Programme on Energy from Urban, Industrial, Agricultural Wastes/Residues

- The objective of the program is to support the setting up of waste-to-energy projects for generation of biogas/bioCNG/power/producer or syngas from urban, industrial, and agricultural wastes/residues. The central financial assistance (CFA) for setting up waste-to-energy plant is as follows:

BioCNG/Enriched Biogas/Compressed Biogas

a) Rs 4 crore per 4800 kg/day (for BioCNG generation from new biogas plant)

b) Rs 3 crore per 4800 kg/day (for BioCNG generation from existing Biogas plant)
(Maximum CFA of Rs 10 crore/project for both cases.)

Is the biomass wet or dry? Wet biomass with 50% water content has less than half the heating value of dry biomass.

Does the biomass contain many plant nutrients? A high ratio of nutrients such as alkalis, phosphorous, and nitrogen makes it difficult to use it for direct combustion. They should rather be recycled to the soil after digestion.

What is the distance to the offtake of energy? Long transport distances should be avoided. And finally, of course: what is the economic return an entrepreneur can expect. This will depend on policy incentives to some extent.

In the long term, one thing is sure. As the world consciously tries to move away from fossil fuels to avoid catastrophic heating of the atmosphere, energy generated from renewable sources will become a valuable good. With the kind of agricultural residues that India produces annually, perhaps, good times are awaiting the agriculture sector in India. And it is quite possible that agricultural residues may have the same value as the crops themselves. It's time to embrace the opportunity!

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